Import Data Transform Map

(Spread Sheet)

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TESTING PHASE

1. Introduction:

Testing is one of the important phases in a software project. The main purpose of testing is to check whether the developed project works correctly according to the expected requirements. Testing helps to identify errors, incorrect data, missing data, duplicate records, and other problems before the project is used in a real environment.In this project, "Import Data Using Transform Maps", testing is mainly performed to verify the complete process of importing employee data from an external Excel spreadsheet into the Service Now platform. The employee information is first loaded into an Import Set table. Then, the Transform Map is used to transfer the data from the source table to the Employee Test target table.The testing phase verifies whether the data is loaded correctly, whether the fields are mapped properly, whether records are inserted successfully, and whether existing records are updated correctly. It also checks whether duplicate records are avoided when the same employee data is imported more than once.

The project uses Import Sets and Transform Maps to perform the data migration process. The testing process therefore focuses mainly on data accuracy, field mapping, transformation, record insertion, record updating, and duplicate prevention.

The overall testing process can be represented as:

Excel Spreadsheet → Import Set Table → Transform Map → Employee Test Table → Validation.

2. Objectives of Testing:

The main objective of testing is to make sure that the project performs the expected operations without errors.

The important objectives of testing are:

1. To check whether the Excel spreadsheet data is loaded correctly.
2. To verify whether the Import Set table receives the source data.
3. To verify whether the source fields are mapped correctly with the target fields.
4. To check whether the Transform Map works correctly.
5. To verify whether employee records are inserted into the target table.
6. To check whether existing employee records are updated correctly.
7. To prevent duplicate employee records.
8. To verify the working of the Coalesce option.
9. To check the Transform History for successful and unsuccessful records.
10. To ensure that the final employee data is accurate and consistent.

Testing therefore helps to ensure that the complete import process works properly from beginning to end.

3. Testing of Import Set Table:

The first major testing activity is checking the Import Set table.The Excel spreadsheet contains employee information such as:

- Employee ID
- Employee Name
- Email
- Department

- Location

- The spreadsheet data is loaded into the Employee Import table. The Import Set table acts as a temporary staging area where the imported data is stored before transformation.
- During testing, the Excel file is selected as the source of import. The existing Employee Import table is selected and the required sheet number and header row are provided.
- After submitting the data, the system displays the import status. The testing checks whether the import is completed successfully.

The following points are verified:

- Whether the Excel file is accepted.
- Whether all required rows are imported.
- Whether the column headers are recognized correctly.
- Whether employee data is available in the Import Set table.
- Whether the import state shows completion and success.

The project document specifies that the Excel data is loaded into the existing Employee Import table and submitted for processing. If the data is successfully loaded, the testing process continues to the transformation stage.

4. Testing of Transform Map

- The next important testing activity is testing the Transform Map.
- A Transform Map defines how the data from the source table should be transferred to the target table.

In this project, the Transform Map is named: Sample Spreadsheet Import

The source table is:

- Employee Import

The target table is:

- Employee Test

The fields from the source table are mapped with the corresponding fields in the target table.

For example:

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

The project uses Auto Map Matching Fields and Mapping Assist to map the source and target fields. After mapping, the Transform Map is saved.

Testing is performed to make sure that:

- Source fields are mapped correctly.
- Target fields receive the correct values.
- No important field is left unmapped.
- The correct target table is selected.
- The Transform Map executes successfully.

If the mapping is incorrect, the employee data may be inserted into the wrong fields. Therefore, field mapping is an important part of the testing phase.

5. Testing of Data Transformation:

- After creating and mapping the Transform Map, the transformation process is executed.
- The Transform option is selected and the transformation is started.
- The main purpose of this test is to check whether the imported data is successfully transferred from the Employee Import table to the Employee Test table.
- After transformation, the system displays the transformation status.

The expected result is:

- State: Completed
- Status: Success
- The project document confirms that after transformation, the records are created in the Employee Test table.

The following conditions are checked during testing:

1. Transformation starts successfully.
2. Source records are processed.
3. Target records are created.
4. Employee ID is transferred correctly.
5. Employee Name is transferred correctly.
6. Email is transferred correctly.
7. Department is transferred correctly.
8. Location is transferred correctly.
9. No unexpected data is lost.
10. The transformation finishes successfully.

After successful transformation, the Employee Test table is opened and the records are verified.

6. Testing of Record Insertion:

- Record insertion testing checks whether new employee records are successfully created in the target table.
- For this test, new employee information is added to the Excel spreadsheet. The file is imported and the Transform Map is executed.
- The newly added employee records should appear in the Employee Test table.
- The project testing process includes adding new Employee IDs and checking the transformation result. In the documented test, four rows were uploaded, including two new employee IDs. These two new records were inserted into the target table.

The expected result is:

New Employee Data → Transform → New Record Created

- This confirms that the system can successfully insert new employee records.

7. Testing of Existing Record Updates:

- Another important test is checking whether existing employee records are updated instead of creating unnecessary duplicate records.

- For this test, an existing Employee ID is used again, but some values such as the employee name or email are changed.

For example:

  - Existing Employee ID remains the same.
  - Employee Name is changed.
  - Email is changed.

- The updated Excel file is imported again.
- After transformation, the system checks the Employee ID and identifies the existing record. The corresponding employee information is then updated.
- The project test shows that out of four uploaded rows, two were new records and two were existing records with changed values. The new records were inserted and the existing records were updated.
- This testing confirms that the project can handle both new data and modified existing data.

. Testing of Coalesce:

- Coalesce testing is an important part of this project.
- When employee data is imported multiple times, there is a possibility of creating duplicate records. To avoid this problem, Coalesce is enabled in the Transform Map.
- In this project, the Employee ID field is used for identifying existing employee records.
- When Coalesce is enabled, the system checks whether the Employee ID already exists.

If the Employee ID does not exist:

- New Employee ID → Insert New Record

If the Employee ID already exists:

- Existing Employee ID → Update Existing Record
- This prevents duplicate employee records.
- The project document explains that Coalesce is used to prevent duplicate employee records and update existing records with new values.

9. Duplicate Data Testing:

- uplicate testing is performed by importing the same Excel data again.

- The purpose of this test is to make sure that the system does not create duplicate employee records.
- The same spreadsheet is uploaded again using the same Import Set and Transform Map.
- After transformation, the Transform History is checked.

The documented test shows that when the same sheet was uploaded again, four rows were processed and:

- Inserted: 0
- Updated: 0
- Ignored: 4

- This shows that the already processed records were not inserted again.
- Therefore, the duplicate testing confirms that Coalesce helps maintain clean and consistent employee data.

10. Testing TR Using Transform History:

- Transform History is used to verify the result of the transformation.
- It provides information about how the imported records were processed.

The testing checks the number of:

- Inserted records
- Ignored records
- Updated records

- For example, during the project testing, four records were uploaded. Two new records were inserted and two existing records were updated.
- When the same data was uploaded again, the records were ignored because they already existed.
- This makes Transform History useful for verifying the result of each import operation.
- It helps the administrator understand whether the transformation was successful and how the imported data was processed.

This report displays employee information according to location.

The configuration includes:

- Table: Employee Test
- Type: Bar Chart
- Group By: Location
- Aggregation: Count

The report is executed and checked to make sure that the employee counts are displayed correctly.

Report 3 – Employee List Report

This report displays the employee records in list format.

The selected columns are:

- Employee ID
- Employee Name
- Email
- Department
- Location

The report is executed and verified to ensure that the required employee information is displayed correctly.

13. Dashboard Testing:

- The final testing activity includes checking the dashboard.
- The project creates an Employee Analytics Dashboard and adds the three reports to it.

The dashboard contains:

1. Employees by Department
2. Employees by Location
3. Employee List Report

- The purpose of dashboard testing is to verify whether all reports are added successfully and whether the information is displayed properly.

11. Validation of Employee Test Table:

- After the transformation is completed, the Employee Test table is opened.
- The purpose of validation is to compare the imported information with the original Excel data.

The following fields are checked:

- Employee ID
- Employee Name
- Email
- Department
- Location

- The values in the Employee Test table should match the corresponding values in the source spreadsheet.
- The records are also rearranged using the Personalized List Columns option so that the required fields can be viewed clearly.
- The project document states that after successful import, all records are created in the Employee Test table and the fields can be arranged for verification.

12. Testing of Reports:

Reports are also created using the Employee Test table.

Three reports are created:

Report 1 – Employees by Department

This report displays employee information based on department.

The report uses:

- Table: Employee Test
- Type: Pie Chart
- Group By: Department
- Aggregation: Count

This report is tested to verify whether employees are correctly grouped according to their departments.

Report 2 – Employees by Location

records can be inserted and existing employee records can be updated. The use of Coalesce prevents duplicate records when the same employee data is imported repeatedly. Transform History helps verify the number of inserted, updated, and ignored records. Reports and dashboards are also tested to provide a clear view of employee information. Overall, the testing phase confirms the accuracy, consistency, and successful operation of the employee data import process in Service Now. The project therefore provides a structured method for importing and managing employee data using Import Sets, Transform Maps, Coalesce, Reports, and Dashboards.ss

- The project document confirms that all three reports are added successfully to the dashboard.
- The dashboard provides a centralized view of employee information and helps administrators monitor employee data.

14. Test Results:

The testing performed in the project gives the following results:

Test Case	Expected Result	Result
Excel Data Imported	Data should load successfully	Pass
Import Set Testing	Source data should be available	Pass
Field Mapping	Fields should map correctly	Pass
Transform Testing	Transformation should complete	Pass
New Record Testing	New employee should be inserted	Pass
Update Testing	Existing employees should be updated	Pass
Coalesce Testing	Duplicate records should be avoided	Pass
Duplicate Testing	Same records should not be inserted again	Pass
Transform History	Inserted/update/Ignored status should be visible	Pass
Report testing	Reports should display employee data	Pass
Dashboard Testing	Reports should appear in dashboard	Pass

15. Conclusion of Testing Phase:

The testing phase confirms that the Import Data Using Transform Maps project works correctly.The project successfully tests the complete employee data import process. The Excel data is loaded into the Import Set table, mapped using the Transform Map, and transferred to the Employee Test table. The testing also confirms that new employee